A rational counterexample to infinite log-concavity

Complete solution of MathDB 347493

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Task 1. Resolve the infinite log-concavity conjecture. For $n \geq 0$, put

$$f_n(z) = \sum_{j=0}^{n} \binom{n}{j} z^{\binom{j}{2}}, \qquad \mathcal{L}(a)_n = a_n^2 - a_{n-1}a_{n+1}.$$

Dilcher and Ulas conjectured that $(f_n(z))_{n\geq 0}$ is infinitely log-concave for every $0 \leq z \leq 1$. This is false: at $z = 3/10$, the sixth iterate satisfies

$$\mathcal{L}^6(f)_6(3/10) < -\frac{63}{10^{36}} < 0. \tag{1}$$

Only $f_0, \ldots, f_{12}$ enter this counterexample, so no convention about negative indices is needed.

A finite exact certificate. Define rational numbers

$$a_{0,n} = \sum_{j=0}^{n} \binom{n}{j} (3/10)^{\binom{j}{2}} \quad (0 \leq n \leq 12), \qquad a_{r+1,n} = a_{r,n}^2 - a_{r,n-1}a_{r,n+1}.$$

At level r we retain $r \leq n \leq 12 - r$. By induction, $a_{r,n} = \mathcal{L}^r(f)_n(3/10)$. Exact rational calculation gives the following strict bounds:

n	5	6	7
lower bound for $10^{18}a_{5,n}$	8	44	261
upper bound for $10^{18}a_{5,n}$	9	45	262

In particular, the three entries are positive, and hence

$$a_{6,6} = a_{5,6}^2 - a_{5,5}a_{5,7} < \frac{45^2 - 8 \cdot 261}{10^{36}} = -\frac{63}{10^{36}}.$$

This proves (1).

For completeness, the exact calculation establishing every bound in the table is reproduced below. This is a finite rational-arithmetic certificate; the code uses only Python's standard library, with no floating point operations.

```
from fractions import Fraction as F
from math import comb

q = F(3, 10)
a = [sum((F(comb(n,j))*q**comb(j,2)
          for j in range(n+1)), F(0))
     for n in range(13)]
for r in range(5):
    a = [a[i]**2-a[i-1]*a[i+1]
         for i in range(1, len(a)-1)]

for x, lo, hi in zip(a, (8,44,261), (9,45,262)):
    assert lo < 10**18*x < hi
assert a[1]**2-a[0]*a[2] < F(-63, 10**36)
```

An integer form of the certificate. The accompanying verification also computes the same entries by a separate integer recurrence. Set

$$P_{0,n} = \sum_{j=0}^{n} \binom{n}{j} 3^{\binom{j}{2}} 10^{\binom{n}{2}-\binom{j}{2}}, \qquad P_{r+1,n} = 10^{2^r} P_{r,n}^2 - P_{r,n-1}P_{r,n+1}.$$

Define $d_{0,n} = \binom{n}{2}$ and, for $r \geq 1$, $d_{r,n} = 2^r \binom{n}{2} + r2^{r-1}$. The identities

$$d_{r+1,n} = 2d_{r,n} + 2^r = d_{r,n-1} + d_{r,n+1}$$

show inductively that $a_{r,n} = P_{r,n}/10^{d_{r,n}}$. All 49 entries in the triangular calculation agree between the two recurrences, and all ten verification groups pass. The terminal rational value and the three preceding values are saved in full in the verification record.

Consequence for the stronger coefficient conjecture. The same source's Conjecture 3.9 asserts positivity of the coefficients of $\mathcal{L}^k(f)_n(z)/(1-z)^{2^k-1}$. At $k = n = 6$ its value at $z = 3/10$ is negative, because the denominator $(7/10)^{63}$ is positive. Thus that stronger assertion fails as well. This is a consequence of the present counterexample, with no additional solve counted here.

Task 2. Determine the parameter set. Characterize those $z \in [0,1]$ for which $(f_n(z))$ is infinitely log-concave.

Task 3. Determine the first failing iteration. For each parameter where infinite log-concavity fails, determine the least iteration and the first index witnessing failure.

Task 4. Study the normalized coefficient signs. Determine when the normalized iterates $\mathcal{L}^k(f)_n(z)/(1-z)^{2^k-1}$ have nonnegative coefficients, and explain the onset of negative coefficients.

Task 5. Find sufficient conditions for binomial transforms. Identify useful hypotheses on a nonnegative sequence (b_j) ensuring infinite log-concavity of $\left(\sum_j \binom{n}{j} b_j\right)_{n\geq 0}$.

References

[1] K. Dilcher and M. Ulas, *Some properties of a class of sparse polynomials*, Journal of Mathematical Analysis and Applications **505** (2022), 125449. Conjectures 3.8 and 3.9, published p. 10; preprint p. 9, arXiv:2008.01480. Published article.